DrainSense: the problem and the solution

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The problem

The people who keep drains clear and respond to floods have no single place that says where flood risk is high, why, and where to send a crew first. The signals exist, but they sit apart:

- **Flood records** show where large floods happened in the past, as a catalogue, not as a forward look.
- **Rainfall and terrain data** show what fell and where water collects, but not what that means for next season.
- **Blocked-drain reports** come from residents one at a time, by phone, message or word of mouth, and are rarely tied to the place's flood risk or to past repairs.

The result is reactive maintenance. Crews are sent where the last complaint came from, not where risk and reports overlap. Nobody can say afterwards whether a repair cut the number of incidents.

Risk scores, where they exist, are often black boxes. A planner cannot see why an area scores high, how good the model is, or which numbers are real and which are filler. A score that cannot be explained or checked is hard to act on.

The same flood risk matters beyond drainage: to farm suppliers planning stock, health teams watching water-borne disease, and insurers and logistics planners pricing exposure. Each would otherwise rebuild the same picture on its own.

What is in it

Five parts, each a page on the desk and a set of API endpoints.

Part	What it answers	How
Risk map	Where is a large flood most likely next year?	A logistic model scores 571 land cells, 5° on a side, for a large flood in 2024 from 2023 rainfall and elevation. Each cell shows the score and the factors that pushed it up or down.
Drainage ledger	What has been reported, inspected and fixed, and where?	Residents' blocked-drain reports, inspections, repairs and incidents, linked to a cell. It counts incidents in the year before and after each repair.
Dispatch plan	Where should a crew go first?	Ranks cells by 1,000 × flood score + 25 per open report + 10 per repeated incident. Every point in the ranking is shown with its source.
Sector flags	What does the risk mean for my sector?	Fixed rules turn the score into attention flags for farm suppliers, health planners, and insurers and logistics. Labelled as derived flags, not separate models.
Flood-impact model	How big was a recorded flood, and what drove it?	A CatBoost model estimates the flood impact index of an event from its size, duration and observed weather, one at a time or 500 at a time. It also returns the exact index formula when area, duration and severity are known.

Everything runs through one API with interactive documentation (Swagger), a published schema, and a Postman collection that tests every endpoint.